DISHA KAMALE

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SUMMARY

I am a postdoctoral researcher at the Autonomous Robotic Manipulation (ARM) Lab, University of Michigan. My research centers on developing planning algorithms for ***long-horizon, complex tasks*** with ***provable safety guarantees***. My ongoing research focus is on generating *reliable* symbolic representations using foundation models for long-horizon planning. I am excited to contribute with my experience in designing novel algorithms for *VLM-guided task and motion planning, planning with infeasibilities and user preferences, and perception-aware sequential decision-making* for ***system-level reasoning, planning, and safety***.

EDUCATION

Lehigh University — *Bethlehem PA, USA*
Ph.D., Mechanical Engineering and Mechanics (GPA: 4.0/4.0) — *May 2025*
Visvesvaraya National Institute of Technology(NIT), Nagpur — India
B.Tech. Mechanical Engineering (GPA: 4.0 - converted using WES) — *2019*

RESEARCH EXPERIENCE

- **Autonomous Robotic Manipulation Lab**, University of Michigan
  Postdoctoral Research Fellow, *Advisor: Prof. Dmitry Berenson* — *August 2025-present*
- **Honda Research Institute (HRI), CA, USA**
  Research Intern *(Patent under review at USPTO)* — *May-August 2024*
  ◦ *Neurosymbolic Path Planning with Provable Guarantees*
- **Autonomous and Intelligent Robotics Lab (AIRLab)**, Lehigh Univerisity
  Graduate Research Assistant, *Advisor: Prof. Cristian-Ioan Vasile* — *2020-2025*
- **Intelligent Manipulation Lab (IntManLab)**, University of Lincoln, Lincoln, UK.
  Research Intern, *Advisor: Prof. Amir Ghalamzan Esfahani* — *2019-2020*
- **Neuro-Robotic and Touch Lab**, The Biorobotics Institute, Pisa, Italy.
  Summer Research Intern, *Advisor: Prof. Calogero Maria Oddo* — *May-August 2018*
  Neuro-Robotic and Touch Lab, The Biorobotics Institute, Pisa, Italy.
- **Robotics lab - IvLabs**, NIT Nagpur.
  Voluntary Student Researcher, *Advisor: Prof. Shital Chiddarwar* — 2016-2019

TECHNICAL SKILLS

Robotics and Simulation	ROS/ROS2, IsaacSim, CARLA, CoppeliaSim, Gazebo, MuJoCo
Programming and Workflow	Python(proficient), MATLAB(proficient), C/C++(comfortable), Git
AI and Learning	PyTorch(proficient), Tensorflow(comfortable)
Optimization	Numerical Methods, Mixed Integer Programming, Convex
Mathematical Background	Stochastic Calculus, Graph Theory, Risk Analysis, Probability Theory, Reinforcement Learning
Robotics Foundations	Robot Kinematics, Dynamics, Path and Motion Planning, Linear, Non-linear, and Optimal Control

SELECT RESEARCH PROJECTS

VLM-guided Task Planning for Long-horizon Manipulation Tasks — *ongoing*
Key Highlights: ***Foundation Models*** | ***Generative Modeling*** | *Isaacsim* | ***Symbolic Reasoning*** | ***Task Planning*** | *PDDL* | *Closed-loop Planning* |

- Designed a closed-loop, PDDL-based task and motion planning framework in Isaacsim for autonomous engine maintenance tasks.

• Proposed and developed a VLM-guided systematic generation of symbolic abstractions for a given task demonstration to enable reliable synthesis of long-horizon plans.

Perception-aware Planning and Decision-making *2020-25*

Key Highlights: ***Foundation Models*** | *Symbolic Reasoning* | *Discrete Planning* | ***Reactive Control*** | ***Sequential Decision-making*** | ***Semantic Perception Modeling*** | ***Risk-aware Planning***

• Currently developing a semantic perception abstraction leveraging vision-language models, enabling robust environmental reasoning under imperfect perception while providing provable guarantees.
• Proposed, designed, and programmed a novel distance-stratified semantic perception and tracking framework, achieving provably correct control and **boosting safety by over 60%** even under noisy, incomplete perception.
• Designed and deployed a CVaR-based planning framework, resulting in a **29% improvement in controller safety** compared to baseline methods under **perception uncertainty**.

Infeasibility-aware Optimal Planning Frameworks for Multi-robot Teams *2020-25*

Key Highlights: ***Low-latency Planning*** | *A** | *MILP Optimization* | *Human Preferences* | ***Multi-agent Systems*** | ***Control Synthesis*** | ***Safety-critical Control***

• Developed scalable system-level planning frameworks integrating symbolic reasoning, user preferences, and formal guarantees to enable correct, adaptive behavior in multi-agent settings.
• Designed a novel Mixed Integer Linear Programming (MILP) formulation for planning with relaxation of infeasible/conflicting requirements, achieving **globally optimal plans** for teams of **400 robots** in **0.8 seconds** for complex high-level tasks with logical and temporal requirements.
• Created an infeasibility-aware A*-based search algorithm, devising near-optimal solutions for metro city-sized graphs (378,040 nodes, over 1M edges) in 3.8 seconds, **over 2.5x faster** than baseline approaches for **real-time planning**.
• Enhanced Dijkstra's algorithm for **multi-objective planning**, improving computational efficiency and solution quality for high-level mission planning.

Inferring Spatio-temporal Behavior From Time-series Data

Key Highlights: *Decision Trees* | *Supervised Learning* | *Signal Temporal Logic*

○ Developed a learning framework for inferring high-level behavior specifications using labeled time series data using binary decision trees.

Research internship: Intelligent Manipulation Lab (IML), University of Lincoln, UK *June-Dec 2019*
Human-in-the-loop: Haptic-guided Collision Free Manipulation Video | Paper

Key Highlights: ***Control*** | *Arm motion planning* | *Grasp planning* | *Human-robot interaction* | *Real robot experiments* | *Kinematics* | *Rigid body transformations*

• Designed and implemented a human-robot shared control architecture for effective human-in-the-loop manipulation on Franka Panda robot arms, providing informative force cues to guide user motion.
• Validated the system through human subject testing, achieving a **28% reduction in time** to task completion compared to baseline real-time control strategies.

PATENT

• Neuro-symbolic Path Planning. *Under review USPTO*
- Developed **neuro-symbolic path planning framework with transformers** with provable guarantees for autonomous agents, creating safer decision-making systems for embodied agents.

PUBLICATIONS

Journals

• **Disha Kamale**, Dmitry Berenson. **VLM-guided Symbolic Abstractions for Long-horizon Planning.** (*Under review*)

• **Disha Kamale**, Cristian-Ioan Vasile. **Optimal Planning with Relaxation of Temporal Logic specifications** (*Under submission, IJRR*)

• Gustavo A. Cardona, **Disha Kamale**, Cristian-Ioan Vasile. **STL and wSTL control synthesis: A disjunction-centric mixed-integer linear programming approach.** NAHS 2025

• **Disha Kamale**, Xi Yu, Cristian-Ioan Vasile. **A*-based Temporal Logic Path Planning with User Preferences on Relaxed Task Satisfaction.** *(In preparation)*

Refereed Conference Papers

- **Disha Kamale**, Fangrui Guo, Xi Yu, Cristian-Ioan Vasile: *Multi-vehicle Planning for Dynamic Urban Environments with Fast Replanning and Task Adaptation* *Under review*, IFAC World Congress 2026
- Kaier Liang, Gustavo Cardona, **Disha Kamale**, Cristian-Ioan Vasile. *Learning Optimal Signal Temporal Logic Decision Trees for Classification: A Max-Flow MILP Formulation* CDC 2024
- **Disha Kamale**, Cristian-Ioan Vasile: *Optimal Control Synthesis with Relaxed Global Temporal Logic Specifications for Homogeneous Multi-robot Teams* ICRA 2024
- **Disha Kamale**, Sofie Haesaert, Cristian-Ioan Vasile: *Energy-Constrained Active Exploration Under Incremental-Resolution Symbolic Perception* CDC 2023
- **Disha Kamale**, Sofie Haesaert, Cristian-Ioan Vasile: *Cautious Planning with Perception: Implementing Verified Reactive Driving Maneuvers* ICRA 2023
- Guangyi Liu, **Disha Kamale**, Cristian-Ioan Vasile, Nader Motee : *Symbolic Perception Risk in Autonomous Driving* ACC 2023
- Gustavo Cardona, **Disha Kamale**, Cristian-Ioan Vasile: *Mixed Integer Linear Programming Approach for Control Synthesis with Weighted Signal Temporal Logic* Invited to special issue HSCC 2023
- **Disha Kamale**, Eleni Karyofilli, Cristian-Ioan Vasile : *Automata-based Optimal Planning with Relaxed Specifications* IROS 2021
- Soran Parsa*, **Disha Kamale***, Sariah Mghames*, Amir Ghalamzan*, et al. : *Haptic-guided shared control grasping for collision-free manipulation.* Best Paper Finalist CASE 2020

Preprint

Muhammad Arshad Khan*, Max Kenney*, Jack Painter*, **Disha Kamale***, Riza Batista-Navarro, Amir Ghalamzan*. **Natural Language Robot Programming: NLP integrated with autonomous robotic grasping**

Workshop Paper

Disha Kamale*, Sariah Mghames*, Tommaso Pardi, Aravinda Srinivasan, Gerhard Neumann, Amir Masoud Ghalamzan Esfahani : **Abstract - Haptic-guiding to Avoid Collision during Teleoperation** - *Open-Ended Learning for Object Perception and Grasping, IROS 2019*

SELECT AWARDS AND HONORS

- **RSS Pioneer 2026** (30 early-career researchers selected among 168 applicants).
- **NSF CPS Rising Star** 2025 awardee. (Top 30 Ph.D. students selected among 174 applicants).
- **ME Rising Star** 2025 awardee - MIT (30 women ME researchers selected across US).
- Recipient of **Rossin Professional Development Program Fellowship** for academic year 2023-24.
- Recipient of **RCEAS Fellowship**, Rossin College of Engineering and Applied Sciences, Spring 2023.
- **Dean's fellow**, Department of Mechanical Engineering and Mechanics for the academic year 2020-21 - Awarded for academic and research excellence
- **Finalist for the best conference paper award** at CASE 2020.
- **Quarter finalist: DST TIIIC** Texas Instruments India Innovation Challenge Design Contest, 2016.

MENTORING

- University of Michigan
  - Rong Fang ((BS, Mathematics) - conformal prediction-guided safety monitoring using VLMs *Research*
  - Jerry Lo (MS, Robotics)- Safe planning for navigation using VLAs *Research*
  - Jason Zou (BS, Robotics) - Planning semantically safe trajectories using VLAs *Research*
- Lehigh University
  - Multiple teams of undergraduate students for Mountaintop Summer Experience Program *Research*
  - Maria Maragkelli *Research*
  - Eleni Karyofylli *Research*
- University of Lincoln
  - Jack Painter *Capstone Project*
  - Max Kenny *Capstone Project*
- NIT, Nagpur
  - Mentored a project on ATGV: Stair Climbing Robot at IvLabs during Summer mentorship project program, VNIT, India. *Technical*
  - Mentored 20 students for academic years 2017-18 and 2018-19 under the Student Mentorship Programme (SMP), VNIT, India. *Academic and career mentoring*
  - An active member of IvLabs; Conducted multiple IEEE workshops at NIT Nagpur.